Pranav Malhotra

pmalhot4@asu.edu • www.linkedin.com/in/mr-pranav-malhotra/ • +1 929-751-3514

EDUCATION

W.P CAREY SCHOOL OF BUSINESS, ARIZONA STATE UNIVERSITY, TEMPE, ARIZONA Graduated: May 2025
Specialized Master of Science in Artificial Intelligence in Business **GPA: 3.90/4**

IRA A. FULTON SCHOOL OF ENGINEERING, ARIZONA STATE UNIVERSITY, TEMPE, ARIZONA Graduated: May 2022
Bachelor of Science in Engineering, Chemical Engineering **GPA: 3.29/4**

RELEVANT WORK EXPERIENCE

AI Engineer Intern (Volunteer) – Center for AI and Data Analytics Lab, ASU Sep 2025 - Present

- Engineer donor-communication datasets through data cleaning, EDA, and feature extraction (Recency–Frequency–Monetary features, sentiment tags), increasing data consistency by about 22 % to date for fine-tuning.
- Fine-tune open-source Hugging Face transformers (FLAN-T5, BERT) using LoRA adapters and prompt-optimization workflows in PyTorch, achieving early improvements of ≈ 17 % in relevance and ≈ 21 % in F1-score on validation data.
- Implement a Retrieval-Augmented Generation (RAG) pipeline with FAISS vector search and Lang Chain, enabling the model to reference verified nonprofit data and showing ≈ 25 % higher factual accuracy in preliminary evaluations.

Machine Learning Engineer Intern (Volunteer) – Professional Training Program, EPICS at ASU, Tempe, AZ Feb 2023 - May 2023

- Productionized an open-source EMO-AffectNet pipeline—integrated Retina Face detection, ResNet50 (VGGFace2) features, and LSTM sequence modelling, processing ≈ 300 frames/min with 2.5 s feedback.
- Piloted CNN-LSTM architecture refinement with dropout regularization and multi-dataset training (AffectNet, RAVDESS, CREMA-D), achieving 66.5% accuracy across 7 emotion classes and improving minority class stability.
- Diagnosed and restructured TorchScript-OpenCV pipeline, reducing model conditioning and inference time by 35%, enabling real-time feedback within 2.5s per segment.

ACADEMIC PROJECTS

Capstone: Threat Intelligence for Arabic-English Multilingual NLP Models (Corporate-Sponsored Koat.AI) Spring 2025

- Led an interdisciplinary team of 5 to fine-tune open-source multilingual transformers (ClapAI's RoBERTa Large, Microsoft's DeBERTaV3, Davlan's mBERT) on synthetically labeled English/Arabic Twitter data for sentiment analysis, emotion detection, and named entity recognition, boosting model performance by approximately 20% across all tasks.
- Applied LoRA adapters for memory-efficient fine-tuning, reducing checkpoint size to under 500M parameters and achieving latency of less than 10 milliseconds on single-GPU Koat.ai servers.
- Containerized and sandbox-deployed the models on AWS EKS (Docker + Kubernetes) to enable real-time analysis of 100K+ English/Arabic tweets in Koat.ai's Scout platform, sustaining < 150 ms end-to-end latency.

Deep Learning & NLP-Based Recommendation System for Yelp Spring 2025

- Led a cross-functional team of four to develop a Yelp-based recommender system, blending sentiment analysis with SVM and Naïve Bayes for Count Vectorizer and TF-IDF, where SVM + TF-IDF yielded an F1 score of 95%.
- Spearheaded TensorFlow/Keras sentiment classification recommendation pipeline leveraging GRU/LSTM with a 100D GloVe embedding, while interfacing AWS S3 for accelerating NLP preprocessing surpassing 94.59% accuracy.
- Orchestrated a content-based filtering approach with BERTopic on 10K reviews, refining 30 topics to 15 with >80% confidence and embedding Sentence Transformer (all-MiniLM-L6-v2, >0.85 confidence).

Machine Learning Driven Residential Real Estate Price Predictor Fall 2024

- Orchestrated a team of four to perform EDA on 40K+ real estate records—handling missing values, removing extreme outliers, normalizing features, and reducing 62 raw attributes to 7 high-impact predictors for scalable model training.
- Deployed Supervised Learning XGBoost regression model, outperforming Linear Regression and Random Forest, using scikit-learn and the XGBoost library with grid search and cross-validation, achieving MAPE < 2% and R^2 > 0.90.
- Implemented three-tier market segmentation (entry, mid, luxury) and spatial analysis (Kepler.gl) to address heterogeneity, reducing luxury-segment MAPE from ~17.5% to ~4% and boosting R^2 from 0.68 to >0.92.

SKILLS

Programming Tools/Skills: Python, SQL, TensorFlow/Keras, NLP, Computer Vision, CNN, Scikit-learn, Keras, Docker, Git, AWS, Pandas, PyTorch, Machine Learning, Deep Learning, Natural Language Processing.
Certifications: Machine Learning Specialization - Coursera Jan 2025

ACHIEVEMENTS AND SCHOLARSHIPS

Dean's List Fall & Spring 2020 – Certificate of Merit – Ira. A Fulton School of Engineering, ASU May 2020 - Jan 2021
New American Scholarship & Summer Scholarship – Finance and Scholarship Services, ASU May 2019 - May 2020
Certificate of Merit (71st / 1,01,104 Rank) – Himachal Pradesh Board of School Education (HPBOSE) April 2016